

00SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 30%

QUESTIONS: 3

Total Marks:30

Annexure C

Case study

Criteria	Understandin g of Case (2.5)	Application of Concepts (2.5)	Analysi s (2)	Solution Design (2)	Clarity (1)	Total(10)
<i>Weightag e</i>	25%	25%	20%	20%	10%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						

Code of Conduct:

*I M Meghanadh(192325026) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

M Meghanadh

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding ; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ANNEXURE A

Case Study:

1. A shepherd boy gets bored tending the town's flock. To have some fun he cries out, "Wolf!" even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them.

One night, the shepherd boy sees a real wolf approaching the flock and calls out, "Wolf!" The villagers refuse to be fooled again and stay in their houses. The hungry wolf turns the flock into lamb chops. The town goes hungry. Panic ensues. Identify TP, FP, FN, TN

Q1. Shepherd Boy Case – Identify TP, FP, FN, TN

In this case, consider:

- **Positive = Wolf is present**
- **Negative = Wolf is absent**
- **Actual = whether a wolf is really present**
- **Prediction = whether the shepherd boy/villagers report a wolf**

The case describes two situations: false alarms when no wolf is present and a real wolf attack when the villagers ignore the warning.

Situation	Actual	Predicted	Classification
Boy cries "Wolf!" when there is no wolf	Negative	Positive	False Positive (FP)
Real wolf approaches but villagers don't respond	Positive	Negative	False Negative (FN)
No wolf and no warning	Negative	Negative	True Negative (TN)
Wolf present and correctly reported	Positive	Positive	True Positive (TP)

Answer

- **TP:** Wolf is actually present and the warning correctly identifies it.
- **FP:** No wolf is present, but the boy cries "Wolf!"
- **FN:** A real wolf is present, but the warning is ignored/treated as no wolf.
- **TN:** No wolf is present and no wolf is reported.

Important case-study observation: The major failure in the final incident is a **False Negative**, because the wolf was actually present but the villagers did not respond to the warning.

2. Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat in 50 of the 70 no cat images. Identify TP, FP, FN, TN and Calculate the Precision and Recall and F1 Score .

Q2. Cat Image Classification

The question states that there are **100 images**, of which **30 contain cats** and **70 do not**. The model correctly predicts cats in **25 of the 30 cat images** and correctly predicts no cat in **50 of the 70 non-cat images**.

Step 1: Identify TP, FN, TN and FP

True Positive (TP)

30 images actually contain cats.

The model identifies 25 correctly:

TP = 25

False Negative (FN)

Out of 30 actual cat images, 25 were correctly identified.

FN = 5

True Negative (TN)

There are 70 images without cats.

The model correctly identifies 50:

TN = 50

False Positive (FP)

The remaining 20 non-cat images are incorrectly predicted as containing cats:

FP = 20

Confusion Matrix

	Actual Cat	Actual No Cat
Predicted Cat	TP = 25	FP = 20
Predicted No Cat	FN = 5	TN = 50
Total	30	70

Step 2: Precision

Precision measures how many images predicted as cats were actually cats.

Precision = 55.56%

Step 3: Recall

Recall measures how many actual cat images were correctly identified.

Recall = 83.33%

Step 4: F1 Score

Using the values:

F1 Score = 66.67%

Final Answer

Measure	Value
TP	25
FP	20
FN	5
TN	50
Precision	55.56%
Recall	83.33%
F1 Score	66.67%

Interpretation: The model has relatively high recall, meaning it identifies most cat images, but its precision is lower because it incorrectly labels 20 non-cat images as cats.

3. An email service provider wants to classify emails as Spam or Not Spam using features:

- **Contains “Offer” (Yes/No)**
- **Contains “Free” (Yes/No)**
- **Email Length (Short/Long)**

Sample Dataset

Offer	Free	Length	Class
Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

Questions

1. **Calculate prior probabilities:**
 - **P(Spam), P(Not Spam)**
2. **Compute conditional probabilities for each feature.**
3. **Using Naïve Bayes, classify:**
 - **(Offer=Yes, Free=No, Length=Short)**

4. **Compare results using WEKA (Naïve Bayes classifier).**
5. **Discuss the independence assumption in Naïve Bayes**

Q3. Naïve Bayes – Spam Email Classification

The dataset contains six emails with three features: Offer, Free, and Length, and the target class Spam/Not Spam.

Given Dataset

Offer Free Length Class

Yes Yes Short Spam

Yes No Long Not Spam

No Yes Short Spam

No No Long Not Spam

Yes Yes Long Spam

No No Short Not Spam

1. Prior Probabilities

There are 6 total emails.

Spam

Spam emails:

- Row 1
- Row 3
- Row 5

Therefore:

Not Spam

Not Spam emails:

- Row 2
- Row 4
- Row 6

Therefore:

Answer

2. Conditional Probabilities

We need to calculate the probability of each feature value given the class.

A. Offer

Given Spam

Spam records:

Offer Class

Yes Spam

No Spam

Yes Spam

Therefore:

Given Not Spam
 Not Spam records:
 Offer Class
 Yes Not Spam
 No Not Spam
 No Not Spam
 Therefore:

B. Free
 Given Spam
 Spam records have:
 • Yes
 • Yes
 • Yes

Therefore:
 Given Not Spam
 Not Spam records have:
 • No
 • No
 • No

Therefore:

C. Length
 Given Spam
 Spam records:
 • Short
 • Short
 • Long

Therefore:
 Given Not Spam
 Not Spam records:
 • Long
 • Long
 • Short

Therefore:

Conditional Probability Table	
Probability	Value
$P(\text{Offer}=\text{Yes} \mid \text{Spam})$	$2/3$
$P(\text{Offer}=\text{No} \mid \text{Spam})$	$1/3$
$P(\text{Offer}=\text{Yes} \mid \text{Not Spam})$	$1/3$
$P(\text{Offer}=\text{No} \mid \text{Not Spam})$	$2/3$
$P(\text{Free}=\text{Yes} \mid \text{Spam})$	1
$P(\text{Free}=\text{No} \mid \text{Spam})$	0

Probability	Value
$P(\text{Free}=\text{Yes} \mid \text{Not Spam})$	0
$P(\text{Free}=\text{No} \mid \text{Not Spam})$	1
$P(\text{Short} \mid \text{Spam})$	$2/3$
$P(\text{Long} \mid \text{Spam})$	$1/3$
$P(\text{Short} \mid \text{Not Spam})$	$1/3$
$P(\text{Long} \mid \text{Not Spam})$	$2/3$

3. Classify: Offer = Yes, Free = No, Length = Short

The assessment asks to classify this email using Naïve Bayes.

Let:

- Offer = Yes
- Free = No
- Length = Short

Naïve Bayes formula:

Probability of Spam

Substitute:

Therefore:

Probability of Not Spam

Therefore:

Final Classification

Note

The zero probabilities occur because the small dataset contains no Spam email with Free = No, and no Not Spam email with Free = Yes. In a practical Naïve Bayes implementation, Laplace smoothing is normally used to avoid zero probabilities.

4. Comparison with WEKA

The assessment specifically asks for a comparison using the WEKA Naïve Bayes classifier.

For this dataset, the manually calculated Naïve Bayes result is:

WEKA's Naïve Bayes classifier should likewise classify this test instance as Not Spam when using the supplied training data and the corresponding categorical attributes.

A suitable comparison is:

Method	Predicted Class
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Manual Naïve Bayes	Not Spam
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WEKA Naïve Bayes	Not Spam
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The exact WEKA output values, such as classifier accuracy, confusion matrix, or probability display, depend on how the dataset is entered and the evaluation method selected.

5. Independence Assumption in Naïve Bayes

Naïve Bayes assumes that the features are conditionally independent given the class.

For this problem, it assumes:

can be represented as:

In other words, once we know whether an email is Spam or Not Spam, the model treats the presence of "Offer", "Free", and the email length as independent of one another.

Advantage

This assumption makes Naïve Bayes:

- Simple to implement
- Fast to train
- Fast for prediction
- Effective even with relatively small datasets

Limitation

In real emails, these features may not actually be independent. For example, an email containing "Free" may also be more likely to contain "Offer". Therefore, the independence assumption may not always hold.